Prevalence of *CFTR* variants in PID patients with bronchiectasis - an important modifying co-factor

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Abstract

Background: Cystic fibrosis (CF) is one of the most common life limiting autosomal recessive disorders and is caused by genetic defects in the cystic fibrosis transmembrane conductance regulator (*CFTR*) gene. Some of the features of this multisystem disease can be present in primary immunodeficiency (PID).

Objective: We hypothesized that a 'carrier' *CFTR* status might be associated with worse outcome regarding structural lung disease in patients with PID.

Methods: A within-cohort and population level statistical genomic analysis of a large European cohort of PID patients was performed using genome sequence data (NIHR BioResource - Rare Diseases PID cohort).

Results: Carriage of several pathogenic *CFTR* gene variants were increased in PID associated with structural lung damage, compared to PID patients without the structural lung damage. Compared to the general population, p.Phe508del carriage was also enriched in lung-related PID.

Conclusion: Genome sequencing identified cases of CFTR dysfunction in PID, driving an increased susceptibility to infection. Large national genomics services provide an opportunity for precision medicine by interpreting subtle features of genomic diversity when treating traditional Mendelian disorders.

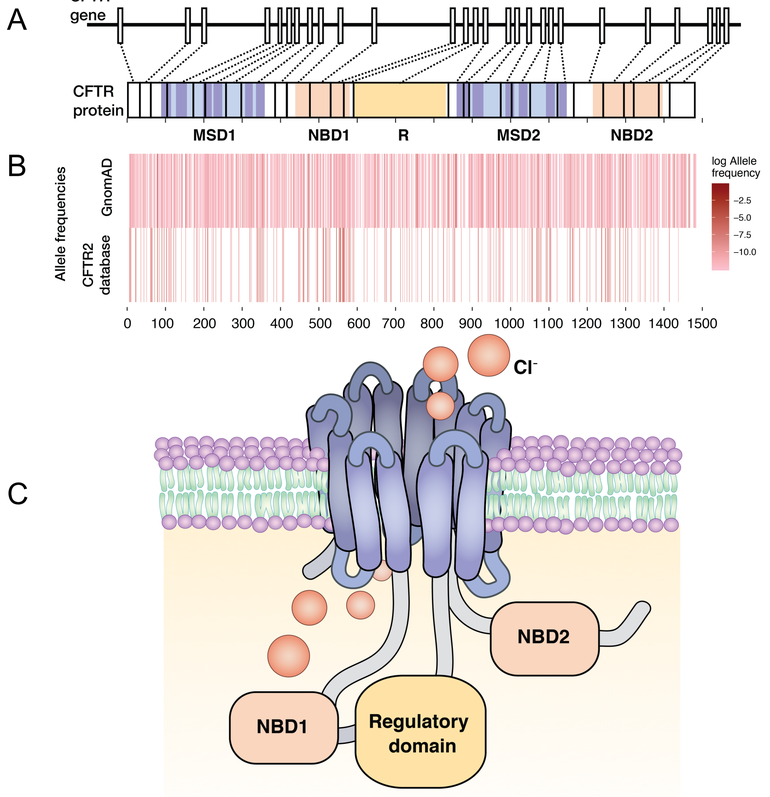


Figure 1: Genomic structure and variant frequency in *CFTR*.

1. Gene structure and variant frequency in *CFTR*

Fig 1(A) The genomic structure of *CFTR* coding exons and functional domains. Black lines indicate exons and functional protein domains indicated by color. **(B)** The allele frequencies of variants reported on GnomAD are shown (population variant frequency) followed by allele frequencies of variants reported in the CFTR2 database (CF-related variant frequency). Combining pathogenic variant frequency and population frequency highlights variants that are most relevant for human health. **(C)** The CFTR protein acting as a trans-membrane channel where it moves chloride ions to outside of the cell. Colors match those as shown on the cDNA structure map. Cl-; chloride ion, MSD1/2; membrane spanning domain, NBD1/2; nucleotide binding domain, R; regulatory domain.

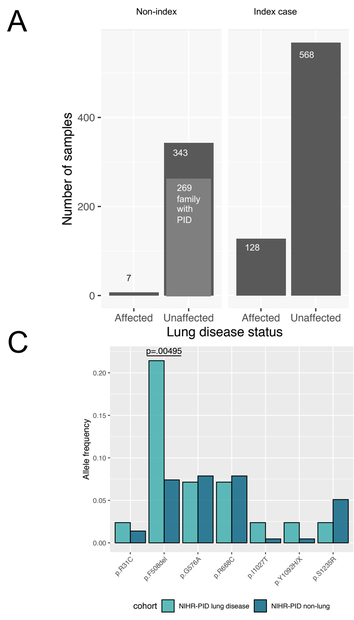


Figure 2: NIHR BR-RD PID candidate variants.

2. NIHR BR-RD PID cohort

Fig 2 (A) The number of PID patients reported as having a lung-related disease status, index cases versus non-index cases recruited either as a healthy relative or with a shared phenotype. Non-index relatives unaffected by lung disease are illustrated as those with (269) and without PID (74). **(B)** Routine analysis of variants produced 3160 total *CFTR* variants within the NIHR BR-RD PID cohort (pink). 118 functional variants passed our filtering criteria (purple). The CFTR2 database had variants affecting 374 coding residues (yellow), and 35 of which were present as candidate variants in our cohort after filtering. Seven of these 35 met our inclusion criteria for subsequent testing.

3. p.Phe508del carriage is enriched in lung-related PID

Fig 2 (C) p.Phe508del was significantly enriched for PID patients with lung-related disease (P = .00495, Padj = .035).

4. Pathogenic *CFTR* variants are enriched for PID-related lung disorder

Fig 2 (D) Enrichment for variants in lung related PID versus the background population. **Padj = .0024, ***Padj <.0001, multiple test correction threshold, P > .007 (Padj > .05). GnomAD; population genetics database, NIHR BR-RD PID; NIHR BioResource - Rare Diseases PID cohort.

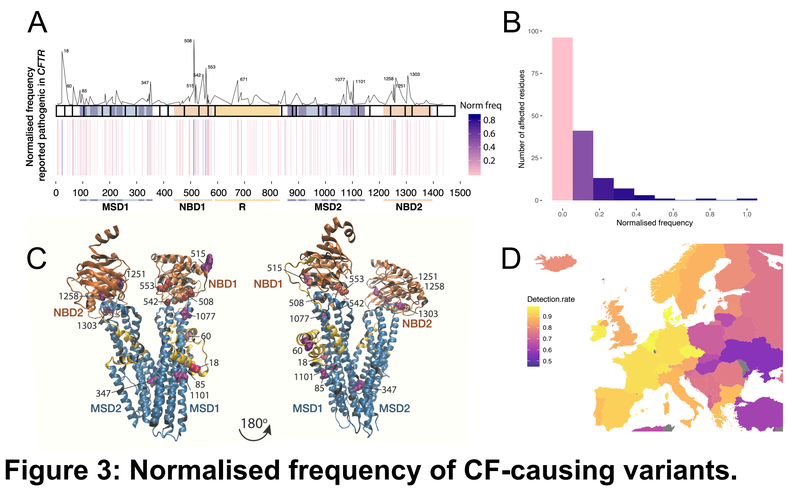


Figure 3: Normalised frequency of CF-causing variants.

5. Normalized pathogenic variant frequency

Fig 3 (A) The normalized frequency of variants (frequency of reported damaging / carriage in the population) is shown by both a line chart top and by color density on bottom. **(B)** Variants were binned according to their clinical relevance. Frequency group >0.5 are most likely to be seen as pathogenic in patients. **(C)** These variants are mapped on CFTR protein structure. **(D)** Detection rates shown across the EU. National screening programs can identify risk modifiers for PID.

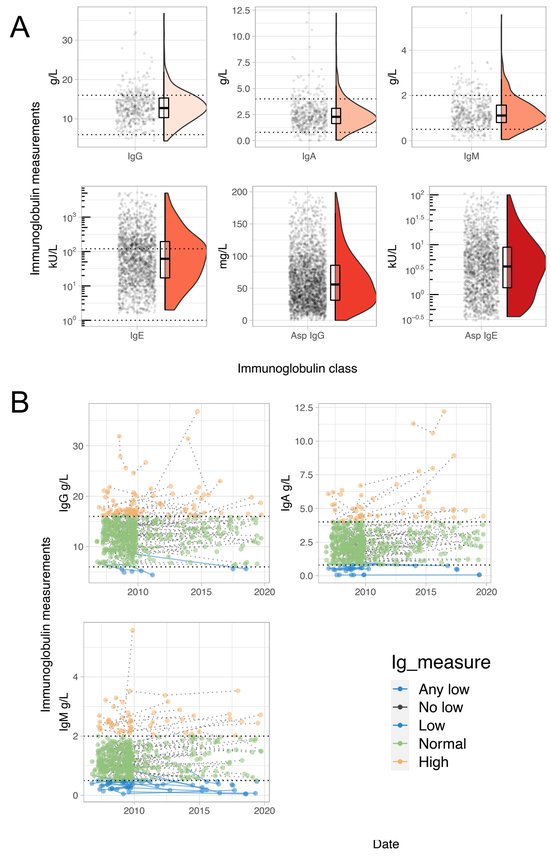


Figure 4: Serum immunoglobulins measurements in 574 confirmed CF cases.

6. Immunoglobulin levels in clinically-confirmed CF

Fig 4 (A) The serum levels for all Ig classes were typically within the normal range. **(B)** We tested outcomes for patients with reduced Ig levels. There were 58 non-transplant deaths and 60 transplant deaths with normal IgA/IgM compared to 16 non-transplant deaths and 9 transplant deaths with low IgA/IgM. Patents found to have reduction in at least one Ig class, had no worse outcome when it comes to survival with or without lung transplant, X2 (1, N = 143) = 1.8, P = 0.2.

Table 1: Additional CFTR variants in NIHR BR-RD PID.

Patient	Member	Variant effect	cDNA	Amino acid	Variant phase	Exon	Phenotype
1	Proband	missense	c.91C>T	p.R31C	A1 [1]	2/27	CID
		inframe deletion	c.1521_1523delCTT	p.IF508I	A2 [1]	11/27	
2	Proband	missense	c.220C>T	p.R74W	A1 [2]	3/27	Autoimmunity
		missense	c.3808G>A	p.D1270N	A1 [2]	23/27	
		missense	c.293A>G	p.Q98R	A2 [3]	4/27	
	Mother	missense	c.293A>G	p.Q98R	A2 [3]	4/27	Unknown
3	Proband	missense	c.220C>T	p.R74W	A1 [2]	3/27	CVID
		missense	c.3808G>A	p.D1270N	A1 [2]	23/27	
4	Proband	missense	c.3705T>G	p.S1235R	A1 [4]	22/27	CVID with Ab defect
		missense	c.2900T>C	p.L967S	A2 [4]	17/27	
	Mother	na	na	na	na	na	CVID with Ab defect
		missense	c.2900T>C	p.L967S	A2	17/27	

7. Newly identified compound heterozygous cases

In addition to assessing the association between heterozygous carriage of known *CFTR* variants and lung disease, we screened the cohort for additional recessive disease. We identified four families with candidate *CFTR* variants. Currently, variants of unknown significance cannot be assessed for their effect on lung function or treatment. Several variants are not considered CF-causing and therefore would likely remain unidentified with a traditional, single-case approach (**Table 1**).

Conclusion

Large national scale sequencing services, exemplified by NIHR BR-RD, allow for in-depth comprehensive statistical analysis, thereby providing an opportunity for precision medicine by interpreting subtle features of genomic diversity when treating traditional Mendelian disorders. This work presents findings of significant importance to understanding the impact of *CFTR* dysfunction in PID.

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